BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 6 — Stoichiometry

ANSWER KEY (Version 1)

Total Marks: 46	**Time Allowed: 1h 45m**	**Version: 1**

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. The empirical formula of glucose ($C_6H_{12}O_6$) is: √ **B — CH_2O** **1 mark**

The **empirical formula** gives the simplest whole-number ratio of atoms. In glucose, C:H:O = 6:12:6 = 1:2:1. So the empirical formula is **CH_2O**. The molecular formula $C_6H_{12}O_6$ is six times the empirical unit.

Q2. The molecular mass of naphthalene ($C_{10}H_8$) is: √ **C — 128 amu** **1 mark**

Molecular mass = (10 × 12) + (8 × 1) = 120 + 8 = **128 amu**. Naphthalene is used in mothballs. Option A (120) forgets the H atoms; option B (108) is incorrect arithmetic.

Q3. The formula mass of $Mg(OH)_2$ is: √ **C — 58 amu** **1 mark**

Formula mass = Mg + 2(O + H) = 24 + 2(16 + 1) = 24 + 34 = **58 amu**. $Mg(OH)_2$ (milk of magnesia) is used to treat acidity. Option D (74) is the formula mass of KCl, a common distractor.

Q4. One mole of any substance contains: √ **B — 6.022×10^{23}** **1 mark**

This experimentally determined number is called **Avogadro's number (N_A)**. One mole of carbon = 6.022×10^{23} atoms; one mole of water = 6.022×10^{23} molecules. Options A and C are off by a power of 10.

Q5. Moles of O_2 molecules in 16 g of oxygen gas: √ **B — 0.5 mol** **1 mark**

Molar mass of O_2 = 2 × 16 = 32 g/mol. Moles = 16 ÷ 32 = **0.5 mol**. Option A (1) confuses 16 g with 1 mol of O atoms, not O_2 molecules — a very common student error.

Q6. Mass of 9.05 moles of ozone (O_3): √ **C — 434.4 g** **1 mark**

Molar mass of O_3 = 3 × 16 = 48 g/mol. Mass = 9.05 × 48 = **434.4 g**. Option A (48 g) is the molar mass of 1 mole only. Ozone forms during thunderstorms from oxygen.

Q7. Atoms in 1.25 moles of zinc: √ **B — 7.53×10^{23}** **1 mark**

Atoms = moles × N_A = 1.25 × 6.022×10^{23} = **7.53×10^{23}** atoms. Zn is used to galvanize steel to prevent corrosion. Option A gives only 1 mole's worth of atoms.

Q8. Formula of aluminium oxide (Al^{3+}, O^{2-}): √ **B — Al_2O_3** **1 mark**

Balance charges: 2(+3) = 3(-2) → **Al_2O_3**. The cation charge becomes the subscript of the anion and vice versa (cross-charge rule). Options C and D give incorrect charge balances.

Q9. Compound with the same empirical and molecular formula: √ **C — H_2O** **1 mark**

For **H_2O**, the ratio H:O = 2:1 cannot be simplified further, so empirical = molecular. H_2O_2 has empirical HO; C_6H_6 has empirical CH; $C_6H_{12}O_6$ has empirical CH_2O.

Q10. Mass of 4 moles of H_2: √ **A — 8.064 g** **1 mark**

Molar mass of H_2 = 2 × 1.008 = 2.016 g/mol. Mass = 4 × 2.016 = **8.064 g**. Option B (4.032 g) is the mass of 2 moles; Option D (1.008 g) is the atomic mass of H alone.

Q11. Balancing chemical equations is done by changing: √ **C — coefficients** **1 mark**

Only **coefficients** (numbers placed before formula units) are changed during balancing. Changing subscripts would alter the identity of compounds. State symbols and element symbols are never changed for balancing purposes.

Q12. Ions that appear on both sides and do not participate are called: √ **B — spectator ions** **1 mark**

Spectator ions appear in both reactant and product sides and are removed to give the **net ionic equation**. In HCl + NaOH → NaCl + H_2O, Na^+ and Cl^- are spectator ions.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) Define empirical formula and molecular formula. Give one example of each. **3 marks**

Empirical formula is the chemical formula that gives the **simplest integer ratio** of the atoms of each element in a compound. **1 mark**

Example: Hydrogen peroxide has one H atom for every O atom, so its empirical formula is **HO**.

Molecular formula is an expression that specifies the **actual number of atoms** of each element in one molecule of a compound. **1 mark**

Example: Hydrogen peroxide molecular formula is **H_2O_2** — two H atoms and two O atoms per molecule. **1 mark**

OR

What is a mole? State Avogadro's number and name three types of representative particles. **3 marks**

A **mole** is the amount of substance that contains as many particles as there are atoms in exactly 12 g of carbon-12. It is equal to **6.022×10^{23}** representative particles. **1 mark**

Avogadro's number (N_A) = 6.022×10^{23} — this experimentally determined constant gives the number of particles in one mole of any substance. **1 mark**

Three types of representative particles: **(i) atoms** (e.g. iron atoms), **(ii) molecules** (e.g. water molecules), **(iii) formula units** (e.g. NaCl). Ions also count as representative particles. **1 mark**

2(ii) Calculate the molecular mass of glucose $C_6H_{12}O_6$. (C=12, H=1.008, O=16) **3 marks**

Molecular mass = sum of atomic masses of all atoms in the molecule. **1 mark**

```
Molecular mass of C₆H₁₂O₆ = 6(C) + 12(H) + 6(O) = 6(12.00) + 12(1.008) + 6(16.00) = 72.00 + 12.096 + 96.00 = 180.096 amu [1 mark]
```

Therefore, molecular mass of glucose = **180.096 amu**. Molar mass = **180.096 g/mol**. **1 mark**

OR

Calculate the formula mass of $CuSO_4 \cdot 5H_2O$. (Cu=63.5, S=32, O=16, H=1) **3 marks**

$CuSO_4 \cdot 5H_2O$ is copper sulphate pentahydrate — an ionic compound with crystal water. **1 mark**

```
Formula mass of CuSO₄·5H₂O Atoms: 1Cu + 1S + 4O (sulphate) + 5×(2H + 1O) (water) = 63.5 + 32 + 4(16) + 5(2×1 + 16) = 63.5 + 32 + 64 + 5(18) = 63.5 + 32 + 64 + 90 = 249.5 amu [2 marks]
```

Formula mass of $CuSO_4 \cdot 5H_2O$ = **249.5 amu**.

2(iii) Differentiate between gram atomic mass, gram molecular mass, and gram formula mass. **3 marks**

Term	Definition	Example
Gram atomic mass	Atomic mass of an element expressed in grams; represents 1 mole of atoms of an element. **1 mk**	C = 12 g; Na = 23 g; Zn = 63.54 g
Gram molecular mass	Molecular mass of a compound expressed in grams; represents 1 mole of molecules. **1 mk**	H_2O = 18.016 g; $C_6H_{12}O_6$ = 180.096 g
Gram formula mass	Formula mass of an ionic compound expressed in grams; represents 1 mole of formula units. **1 mk**	NaCl = 58.5 g; KCl = 74.5 g

All three quantities represent **molar mass** (mass of one mole of a substance in grams).

OR

Differentiate between molecular formula and empirical formula. Which two quantities are needed to find the molecular formula from the empirical formula? **3 marks**

Empirical Formula	Molecular Formula
Gives the **simplest integer ratio** of atoms in a compound. **1 mk**	Gives the **actual number of atoms** in one molecule. **1 mk**
Example: HO (for H_2O_2); CH (for benzene)	Example: H_2O_2; C_6H_6

Two quantities needed: **(1) the empirical formula mass** and **(2) the actual molecular mass** (molar mass from experiment). The ratio gives the multiplier *n* = Molecular mass ÷ Empirical formula mass. **1 mark**

2(iv) Calculate the mass of 0.25 moles of CO_2. (C=12, O=16) **3 marks**

CO_2 is carbon dioxide, a molecular compound. **1 mark**

```
Molar mass of CO₂ = 12 + 2(16) = 12 + 32 = 44 g/mol Formula: mass = moles × molar mass [1 mark] mass = 0.25 × 44 = 11 g [1 mark]
```

Mass of 0.25 moles of CO_2 = **11 g**.

OR

A balloon is filled with 5 g of hydrogen gas (H_2). Calculate the number of moles. (H=1.008) **3 marks**

```
Molar mass of H₂ = 2 × 1.008 = 2.016 g/mol [1 mark] Formula: moles = mass ÷ molar mass [1 mark] moles = 5 ÷ 2.016 = 2.48 moles of H₂ [1 mark]
```

Number of moles of H_2 in 5 g = **2.48 mol**.

2(v) How many molecules are present in 0.5 moles of methane (CH_4)? **3 marks**

Methane (CH_4) is a molecular compound — its representative particles are **molecules**. **1 mark**

```
Formula: particles = moles × Nₐ [1 mark] = 0.5 × 6.022 × 10²³ = 3.011 × 10²³ molecules [1 mark]
```

Molecules in 0.5 mol CH_4 = **3.011×10^{23} molecules**.

OR

A sample contains 3.011×10^{23} Ti atoms. Calculate the number of moles of Ti. **3 marks**

Titanium (Ti) is a corrosion-resistant metal used in rockets and jet engines. **1 mark**

```
Formula: moles = particles ÷ Nₐ [1 mark] = 3.011 × 10²³ ÷ 6.022 × 10²³ = 0.5 moles of Ti [1 mark]
```

Moles of Ti = **0.5 mol**.

2(vi) Write chemical formulas of: (a) NaCl, (b) Mg_3N_2, (c) K_2SO_4. **3 marks**

For binary ionic compounds: write cation first, balance charges using smallest coefficients, write as subscripts.

Compound	Ions	Charge balance	Formula
(a) Sodium chloride	Na^+, Cl^-	1(+1) = 1(−1)	**NaCl** **1 mk**
(b) Magnesium nitride	Mg^{2+}, N^{3-}	3(+2) = 2(−3) → +6 = −6	**Mg_3N_2** **1 mk**
(c) Potassium sulphate	K^+, SO_4^{2-}	2(+1) = 1(−2)	**K_2SO_4** **1 mk**

OR

Write the structural and molecular formula of n-butane. Also convert CH_3-CH_2-OH to molecular formula. **3 marks**

Structural formula shows the arrangement of atoms; **molecular formula** shows the count of each atom. **1 mark**

```
n-Butane structural: CH₃-CH₂-CH₂-CH₃ Count atoms: C = 4, H = 10 Molecular formula: C₄H₁₀ [1 mark] CH₃-CH₂-OH: Count: C=2, H=5, O=1 → but H count: CH₃ = 3H, CH₂ = 2H, OH = 1H → total H = 6 Molecular formula: C₂H₆O (or C₂H₅OH) [1 mark]
```

2(vii) Balance: $CH_{4(g)} + O_{2(g)} \rightarrow CO_{2(g)} + H_2O_{(l)}$ **3 marks**

Balancing chemical equations: use **coefficients** only; never change subscripts. **1 mark**

```
Unbalanced: CH₄ + O₂ → CO₂ + H₂O Step 1: Balance H — put 2 before H₂O: CH₄ + O₂ → CO₂ + 2H₂O Step 2: Balance O — left: 2O; right: 2+2=4O → put 2 before O₂: CH₄ + 2O₂ → CO₂ + 2H₂O [1 mark] Step 3: Check all atoms: C: 1=1 ✓ H: 4=4 ✓ O: 4=4 ✓ Balanced equation: CH₄(g) + 2O₂(g) → CO₂(g) + 2H₂O(l) [1 mark]
```

OR

Balance: $Na_{(s)} + H_2O_{(l)} \rightarrow NaOH_{(aq)} + H_{2(g)}$ **3 marks**

```
Unbalanced: Na + H₂O → NaOH + H₂ Count atoms: Left: Na=1, H=2, O=1 Right: Na=1, O=1, H=2+1=3 → unbalanced H [1 mark] Step 1: Put 2 before NaOH and 2 before Na: 2Na + 2H₂O → 2NaOH + H₂ Check: Na=2=2 ✓ O=2=2 ✓ H=4=4 ✓ [1 mark] Balanced: 2Na(s) + 2H₂O(l) → 2NaOH(aq) + H₂(g) [1 mark]
```

2(viii) How many moles and what mass in 3.011×10^{22} molecules of formaldehyde (CH_2O)? (C=12, H=1, O=16) 3 marks

Formaldehyde is used to preserve dead animals. Its formula is CH_2O. **1 mark**

```
Moles = particles ÷ Nₐ = 3.011×10²² ÷ 6.022×10²³ = 0.05 moles [1 mark] Molar mass of CH₂O = 12 + 2(1) + 16 = 30 g/mol Mass = moles × molar mass = 0.05 × 30 = 1.5 g [1 mark]
```

0.05 moles; mass = **1.5 g**.

OR

Calculate the number of atoms in 0.2 moles of aluminium (Al). 3 marks

Aluminium is used as a wrapper in food industries. Al is an element — its representative particles are **atoms**. **1 mark**

```
Atoms = moles × Nₐ [1 mark] = 0.2 × 6.022 × 10²³ = 1.2044 × 10²³ atoms [1 mark]
```

Atoms in 0.2 mol Al = **1.2044×10^{23} atoms**.

2(ix) Write the steps for writing the formula of a binary ionic compound. Apply to Al^{3+}, O^{2-}. 3 marks

Steps for writing chemical formula of a binary ionic compound: **1 mark**

1. **Step 1:** Write symbols for the cation first, then the anion with their charges.
2. **Step 2:** Balance the charges using the smallest coefficients so total positive charge = total negative charge.
3. **Step 3:** Write the coefficients as subscripts for each ion.
4. **Step 4:** Write the formula, leaving out charges and subscripts of 1.

Application — Aluminium oxide: **2 marks**

```
Write ions: Al³⁺ O²⁻ Balance: 2(+3) = 3(-2) → +6 = -6 ✓ Subscripts: Al₂O₃ Formula: Al₂O₃
```

OR

Define ionic equation. Write the ionic equation for $HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$. 3 marks

An **ionic equation** is a chemical equation in which substances dissolved in water are written as individual ions, and **spectator ions** (common ions on both sides) are removed to give the net ionic equation. **1 mark**

```
Molecular: HCl(aq) + NaOH(aq) → NaCl(aq) + H₂O(l) Expanded ionic (write all aqueous as ions): [1 mark] H⁺(aq) + Cl⁻(aq) + Na⁺(aq) + OH⁻(aq) → Na⁺(aq) + Cl⁻(aq) + H₂O(l) Remove spectator ions (Na⁺ and Cl⁻): [1 mark] H⁺(aq) + OH⁻(aq) → H₂O(l)
```

2(x) Calculate moles in: (a) 15 g of NaCl (b) 40 g of sulphur (S). (Na=23, Cl=35.5, S=32) 3 marks

Formula: moles = mass ÷ molar mass. **1 mark**

```
(a) Molar mass of NaCl = 23 + 35.5 = 58.5 g/mol moles = 15 ÷ 58.5 = 0.256 mol ≈ 0.26 mol [1 mark] (b) Molar mass of S = 32 g/mol moles = 40 ÷ 32 = 1.25 mol [1 mark]
```

OR

Calculate mass of: (a) 1.2 moles of K (b) 0.25 moles of steam (H_2O). (K=39, H=1.008, O=16) 3 marks

Formula: mass = moles × molar mass. **1 mark**

```
(a) Molar mass of K = 39 g/mol mass = 1.2 × 39 = 46.8 g [1 mark] (b) Molar mass of H₂O = 2(1.008) + 16 = 18.016 g/mol mass = 0.25 × 18.016 = 4.504 g ≈ 4.5 g [1 mark]
```

2(xi) Aspirin ($C_9H_8O_4$): (a) Empirical formula (b) Molecular mass. (C=12, H=1, O=16) **3 marks**

Aspirin is used as a mild painkiller. It contains 9C, 8H, 4O atoms. **1 mark**

```
(a) Empirical formula: Ratio C:H:O = 9:8:4 GCD of 9,8,4 = 1 → already in simplest ratio Empirical formula = C₉H₈O₄ (same as molecular) [1 mark] (b) Molecular mass: = 9(12) + 8(1) + 4(16) = 108 + 8 + 64 = 180 amu [1 mark]
```

OR

Caffeine ($C_8H_{10}N_4O_2$): (a) Empirical formula (b) Molecular mass. (C=12, H=1, N=14, O=16) **3 marks**

Caffeine is found in tea and coffee. Its formula is $C_8H_{10}N_4O_2$. **1 mark**

```
(a) Ratio C:H:N:O = 8:10:4:2 Divide by GCD = 2: 4:5:2:1 Empirical formula = C₄H₅N₂O [1 mark] (b) Molecular mass: = 8(12) + 10(1) + 4(14) + 2(16) = 96 + 10 + 56 + 32 = 194 amu [1 mark]
```

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i)(a) Define molecular mass and formula mass with examples. **2 marks**

Molecular mass is the sum of the atomic masses of all atoms present in one molecule of a compound. It is used for covalent/molecular compounds. **1 mark**
Example: Molecular mass of H_2O = 2(1.008) + 16 = **18.016 amu**.

Formula mass is the sum of atomic masses of all atoms in the formula unit of an ionic compound. Ionic compounds exist as arrays of ions, not discrete molecules, so formula mass is used instead of molecular mass. **1 mark**
Example: Formula mass of NaCl = 23 + 35.5 = **58.5 amu**.

3(i)(b) Calculate formula mass of $KClO_3$ and $NaHCO_3$. (K=39, Cl=35.5, O=16, Na=23, H=1, C=12) **3 marks**

```
Formula mass of KClO₃: = K + Cl + 3(O) = 39 + 35.5 + 3(16) = 39 + 35.5 + 48 = 122.5 amu [1.5 marks] Formula mass of NaHCO₃ (baking soda): = Na + H + C + 3(O) = 23 + 1 + 12 + 3(16) = 23 + 1 + 12 + 48 = 84 amu [1.5 marks]
```

$KClO_3$ = **122.5 amu**; $NaHCO_3$ = **84 amu**.

OR

3(i) OR (a) Empirical vs molecular formula with benzene and H_2O_2 examples. **2 marks**

The **empirical formula** gives the simplest integer ratio of atoms; the **molecular formula** gives the actual number of atoms per molecule. **1 mark**

Compound	Molecular Formula	Empirical Formula	Relationship
Benzene	C_6H_6	CH	Molecular = 6 × Empirical
Hydrogen peroxide	H_2O_2	HO	Molecular = 2 × Empirical

Each benzene molecule has 6 C and 6 H atoms; ratio is 1:1 → empirical CH. **1 mark**

3(i) OR (b) Empirical formulas and molecular masses of Aspirin, Vinegar (CH_3COOH), TNT. **3 marks**

```
(i) Aspirin C₉H₈O₄: Ratio 9:8:4 → GCD=1 → Empirical: C₉H₈O₄ Mol. mass = 9(12)+8(1)+4(16) = 108+8+64 = 180 amu [1 mk] (ii) Vinegar: CH₃COOH = C₂H₄O₂ Ratio C:H:O = 2:4:2 → divide by 2 → Empirical: CH₂O Mol. mass = 2(12)+4(1)+2(16) = 24+4+32 = 60 amu [1 mk] (iii) TNT C₇H₅N₃O₆: Ratio 7:5:3:6 → GCD=1 → Empirical: C₇H₅N₃O₆ Mol. mass = 7(12)+5(1)+3(14)+6(16) = 84+5+42+96 = 227 amu [1 mk]
```

3(ii)(a) Define mole and Avogadro's number. Differentiate gram atomic, molecular, and formula masses. 3 marks

A **mole** is the amount of substance containing 6.022×10^{23} representative particles. This number is **Avogadro's number (N_A)** — an experimentally determined constant. Mole can also be defined as atomic mass, molecular mass, or formula mass expressed in grams. **1 mark**

Quantity	Represents	Particles	Example
Gram atomic mass **1 mk**	1 mole of atoms of an element	6.022×10^{23} atoms	C=12g, Na=23g
Gram molecular mass **½ mk**	1 mole of molecules of a compound	6.022×10^{23} molecules	H_2O=18.016g
Gram formula mass **½ mk**	1 mole of formula units of ionic compound	6.022×10^{23} formula units	NaCl=58.5g

All three represent **molar mass** — mass of one mole of a substance in grams.

3(ii)(b) Molecules in: (i) 2.5 mol CO_2 (ii) 3.4 mol NH_3. 2 marks

```
(i) CO₂: molecules = 2.5 × 6.022×10²³ = 1.5055×10²⁴ molecules [1 mark] (ii) NH₃: molecules = 3.4 × 6.022×10²³ = 2.048×10²⁴ molecules [1 mark]
```

OR

3(ii) OR (a) Calculate moles in: (i) 2.4 g He (ii) 250 mg C (iii) 40 g S. (He=4, C=12, S=32) 3 marks

```
Formula: moles = mass ÷ molar mass (i) He: moles = 2.4 ÷ 4 = 0.6 mol [1 mark] (ii) C: 250 mg = 0.25 g moles = 0.25 ÷ 12 = 0.0208 mol [1 mark] (iii) S: moles = 40 ÷ 32 = 1.25 mol [1 mark]
```

3(ii) OR (b) Mass of: (i) 75 mol H_2 (ii) 0.15 mol H_2SO_4. (H=1.008, S=32, O=16) 2 marks

```
Formula: mass = moles × molar mass (i) Molar mass H₂ = 2×1.008 = 2.016 g/mol mass = 75 × 2.016 = 151.2 g [1 mark] (ii) Molar mass H₂SO₄ = 2(1.008)+32+4(16) = 2.016+32+64 = 98.016 g/mol mass = 0.15 × 98.016 = 14.70 g [1 mark]
```

3(iii)(a) Steps for balancing a chemical equation and its importance. **2 marks**

A **chemical equation** is the symbolic representation of a chemical reaction showing reactants and products. Balancing is essential because the **Law of Conservation of Mass** states atoms are neither created nor destroyed — the total number of each type of atom must be equal on both sides. **1 mark**

Steps for balancing: **1 mark**

1. Write the unbalanced equation with correct formulas and state symbols.
2. Count atoms of each element on both sides.
3. Balance one element at a time using **coefficients** (not subscripts), starting with the lowest coefficient.
4. Check all atoms are equal on both sides.

3(iii)(b) Balance: (i) $NH_3 \rightarrow N_2+H_2$ (ii) $Al+HCl \rightarrow AlCl_3+H_2$ (iii) $Fe+O_2 \rightarrow Fe_2O_3$ **3 marks**

(i) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$ N: 2=2 ✓ H: 6=6 ✓ [1 mark] (ii) $2Al(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2(g)$ Al: 2=2 ✓ H: 6=6 ✓ Cl: 6=6 ✓ [1 mark] (iii) $4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ Fe: 4=4 ✓ O: 6=6 ✓ [1 mark]

OR

3(iii) OR (a) Define ionic equation and spectator ions. **2 marks**

An **ionic equation** is a chemical equation in which all substances dissolved in water are written as dissociated ions. **1 mark**

Spectator ions are ions that appear on both sides of the ionic equation unchanged — they do not take part in the actual chemical reaction. They are removed from the expanded ionic equation to produce the **net ionic equation**. **1 mark**

3(iii) OR (b) Write ionic equations for the three reactions. **3 marks**

(i) $Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$ Expanded: $Mg(s) + 2H^+(aq) + SO_4^{2-}(aq) \rightarrow Mg^{2+}(aq) + SO_4^{2-}(aq) + H_2(g)$ Remove SO_4^{2-} (spectator): Net ionic: $Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$ [1 mk] (ii) $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$ Expanded: $Ag^++NO_3^-+Na^++Cl^- \rightarrow AgCl(s)+Na^++NO_3^-$ Remove Na^+, NO_3^- (spectators): Net ionic: $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$ [1 mk] (iii) $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$ Expanded: $Zn(s) + 2H^+(aq) + 2Cl^-(aq) \rightarrow Zn^{2+}(aq) + 2Cl^-(aq) + H_2(g)$ Remove $2Cl^-$ (spectators): Net ionic: $Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$ [1 mk]

3(iv)(a) 0.5 g K added to water. $K_{(s)} + H_2O_{(l)} \rightarrow KOH_{(aq)} + H_{2(g)}$. [K=39] 5 marks

(i) Balance the equation: 1 mark

```
Unbalanced: K + H₂O → KOH + H₂ Left: K=1, H=2, O=1 Right: K=1, O=1, H=2+1=3 → unbalanced Put 2 before K, 2 before H₂O, 2 before KOH: 2K(s) + 2H₂O(l) → 2KOH(aq) + H₂(g) Check: K=2=2 ✓ H=4=4 ✓ O=2=2 ✓ ✓
```

(ii) Moles of K: 2 marks

```
Molar mass of K = 39 g/mol [1 mark] moles = mass ÷ molar mass = 0.5 ÷ 39 = 0.01282 mol ≈ 1.28×10⁻² mol [1 mark]
```

(iii) Number of K atoms: 2 marks

```
atoms = moles × Nₐ [1 mark] = 1.28×10⁻² × 6.022×10²³ = 7.71×10²¹ atoms [1 mark]
```

OR

3(iv) OR — $KClO_3$ decomposes to $KCl + O_2$. [K=39, Cl=35.5, O=16] 5 marks

(i) Write and balance the equation: 1 mark

```
Unbalanced: KClO₃ → KCl + O₂ Oxygen: left=3, right=2 → LCM=6 Put 2 before KClO₃, 2 before KCl, 3 before O₂: 2KClO₃(s) → 2KCl(s) + 3O₂(g) Check: K=2=2 ✓ Cl=2=2 ✓ O=6=6 ✓ ✓
```

(ii) Formula mass of $KClO_3$: 2 marks

```
Formula mass = K + Cl + 3(O) [1 mark] = 39 + 35.5 + 3(16) = 39 + 35.5 + 48 = 122.5 amu [1 mark]
```

(iii) Moles of $KClO_3$ in 24.5 g: 2 marks

```
Molar mass of KClO₃ = 122.5 g/mol [1 mark] moles = 24.5 ÷ 122.5 = 0.2 mol [1 mark]
```

0.2 moles of $KClO_3$ are present in 24.5 g.

— End of Answer Key —